Bacteriological Techniques

Simple Staining

A) Positive Staining:-

Methylene Blue Staining (Morphology of Bacteria)

1) Prepare a smear, allow to dry and then fix with gentle heat
2) Apply the stain for 3 minutes
3) Wash with water and blot dry
4) Bacteria stain blue

B) Negative Staining:-

India Ink Method (Morphology of Bacteria)

1) Mix the small quantity of India Ink on slide with the culture or specimen containing bacteria
2) Make a thin film with the help of platinum loop or another glass side
3) Allow to dry
4) Bacteria are seen as clear transparent objects on a dark brown background

Grams Staining

Procedure:-

1) Fix the smear and allow to cool

2) Place the numbered slides on two glass rods over the staining tray with the smears facing upwards

2) Crystal Violet - 1 minute
(Stains bacteria deep violet)
Drain off the solution, rinse with tap water & drain

3) Gram's Iodine Solution – 2 minutes
(Fixes the violet colour more or less strongly in the bacteria)
Drain off the solution, rinse with tap water & drain

4) 95% Ethanol – ½ minute
(Decolourizes certain bacteria when violet stain is not strongly fixed by iodine solution. It does not decolourize other bacteria when the violet stain is strongly fixed by iodine solution)
Drain off the solution, rinse with tap water & drain

5) *Safranin* Solution – 10 seconds
(Restain pink the bacteria decoloured by ethanol. It has no effect on other bacteria which remain dark violet)
Drain off the solution, rinse with tap water & drain & allow to dry in the air
See under oil immersion objective

Gram Positive – Stains dark purple/violet
(Staphylococci, Streptococci, Micrococci, Pneumococci, Enterococci, Diphtheria Bacilli, Anthrax Bacilli)

Gram Negative – Stains pink
(Gonococci, Meningococci, Coliform Bacilli, Salmonella)

Ziehl- Neelsen's Staining

(Acid-Fast Staining)

The Mycobacteria are acid fast i.e. once they are stained red by *Carbol Fuschin* they cannot be decolourized by acid or alcohol

Procedure:-

1) Prepare a smear and heat fix and allow it to cool

2) Place the numbered slides on two glass rods over the staining tray with the smears facing upwards

3) *Carbol Fuschin* Staining – 5 minutes with heat
(Stains all organisms red)

Cover the smear with piece of cut filter paper. Pour *Carbol Fuschin* from a drop bottle on the filter paper till it is completely covered. Dip the cotton wool swab in the methylated spirit, ignite and pass slowly under the slide to heat it. As soon as steam begins to rise, set the timer at 5 minutes. After 2 ½ minutes heat it once again until steam is seen, but do not let it boil. At the end of another 2 ½ minute remove the filter paper. Cool the slide, drain of the stain; wash the slide gently with water until the water runs off is colourless.

4) Decolourization with acid and ethanol – Acid 5 ½ minutes & Alcohol 2 minutes
(Decolourize all organisms and cellular elements except acid fast bacilli)

Cover the slide with 25% Sulphuric Acid. Act it for 5 ½ minutes to remove the colour of *Carbol Fuschin*. Wash the slide in tap water and drain. The smear may now have a faint pink colour. Pour 90% alcohol or methylated spirit on the smear and leave for 2 minutes. Wash with water and drain.

5) Methylene Blue Counterstaining – 10 seconds
(Stains blue all the organisms and elements decolourized by the acid and ethanol, but acid fast bacilli remains red)

Cover the slide with stain. Leave for 10 seconds. Wash well with tap water and drain. Leave to dry in air on a slide rack.

See under oil immersion objective.

Tubercle bacilli are bright red on a blue background, straight or slightly curved in groups of 3-10 bacilli.